

Run 01 | TEA

Global (including India)

IV only

Target Total early-stage entrepreneurial activity; binary 1 vs. 0
Sample 1,342,884 respondents **Coverage** 89 countries; 2015–2022
Positive cases 154,420 (11.50%) **Case time** 37m 06s
Train / val. / test 913,163 / 161,145 / 268,576 (68% / 12% / 20%, stratified)
Predictors (13): Self-efficacy, opportunity perception, fear of failure, knowing entrepreneurs, intention, media coverage, business exit, business angel, established business owner, good career choice, ease of starting a business, high status, social enterprise.

Legatum predictors (13): Prosperity index score, Safety and security, Personal freedom, Governance, Social capital, Investment environment, Enterprise conditions, Market access and infrastructure, Economic quality, Living conditions, Health, Education, Natural environment. Total independent variables: 26.

Controls (0): None. Country and respondent ID are excluded as predictors.

Protocol: Ten CPU models; model seed 20260925; fixed splits by sample/outcome; training-only preprocessing; F1 threshold selected on validation data. Test scores below are unweighted.

Result in brief

Highest observed ROC-AUC: XGBoost (0.9914). Highest F1: Random Forest (0.8656). The all-negative accuracy baseline is 88.50% (full sample).

Model performance on held-out test records

Model	ROC-AUC	PR-AUC	Acc.%	Prec.%	Recall%	F1	Time
XGBoost	0.9914	0.9305	96.81	84.46	88.53	0.8645	18s
Random Forest	0.9913	0.9290	96.84	84.73	88.47	0.8656	2m 21s
CatBoost	0.9912	0.9279	96.77	84.03	88.78	0.8634	1m 39s
H2O GBM	0.9903	0.9178	96.57	81.91	90.06	0.8579	53s
Elastic Net	0.9889	0.9035	96.33	79.99	90.75	0.8503	9m 53s
Lasso	0.9889	0.9033	96.32	79.99	90.71	0.8501	8m 41s
Logistic Regression	0.9888	0.9030	96.39	81.32	89.06	0.8501	1m 24s
Ridge	0.9883	0.8986	96.25	79.74	90.38	0.8473	8m 00s
Decision Tree	0.9859	0.8495	96.33	81.91	87.39	0.8456	10s
Naive Bayes	0.8954	0.4178	85.99	43.68	75.52	0.5535	3m 23s

Models ranked by ROC-AUC; small differences are descriptive, not significance tests. Time includes fit, prediction and output processing. Logistic regression uses a small stabilizing penalty; Lasso/Ridge/Elastic Net use 3-fold training CV.

Random-forest feature importance: top 8

Rank	Predictor	Raw importance	Share%
1	Entrepreneurial intention	78,884.89	55.07
2	Established business owner	27,250.13	19.02
3	Self-efficacy	4,452.97	3.11
4	Knowing entrepreneurs	4,002.40	2.79
5	Education	1,895.43	1.32
6	Safety and security	1,893.33	1.32
7	Living conditions	1,722.51	1.20
8	Opportunity perception	1,447.50	1.01

Share is the percentage of total forest impurity importance across all 51 encoded predictors, not explained variance. Raw values are not comparable across samples. Missing indicators are derived preprocessing features.

Leading Legatum predictors within the same forest

Rank	Legatum predictor	Raw importance	Share%
1	Education	1,895.43	1.32
2	Safety and security	1,893.33	1.32
3	Living conditions	1,722.51	1.20

Legatum subset ranked by raw importance; shares use total forest importance, not the Legatum subtotal.

Source: Completed Objective 2 batch, 28 September 2026; 01_obj2_global_iv_TEA. Scores and importance are taken directly from saved CSV outputs.